

MOHANKUMAR MARIMUTHU

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EDUCATION

Bachelor of Technology in Artificial Intelligence & Data Science

2022 – 2026

Karpagam Academy Of Higher Education

Coimbatore, Tamil Nadu

– CGPA: 8.2

SKILLS

AI/ML

LLM Orchestration, RAG, NLP, Prompt Engineering, Vector Search, TF-IDF, NLI

Languages

Python, C, Java, JavaScript, HTML, CSS, SQL

Frameworks/Tools

PyTorch, TensorFlow, Flask, FastAPI, Docker, Git, AWS, N8N

INTERNSHIPS

Artificial Intelligence Intern

Dec 2023 – Jan 2024

Altitudes Solutions

Coimbatore, Tamil Nadu

- Designed and built a RAG chatbot from scratch (no ML libraries) over a 192K-message dataset, implementing custom TF-IDF retrieval and cosine-similarity topic detection.
- Engineered sliding-window topic segmentation (526 topics, 1,919 checkpoints) and a rule/regex-based persona extraction layer for habits, traits, and communication style.
- Served the pipeline via a Flask REST API with Claude Sonnet 4 as a grounded generator, with a rule-based fallback when no API key is present.

Artificial Intelligence Intern

Oct 2024 – Nov 2024

Prime Solutions

Coimbatore, Tamil Nadu

- Built a conversational recommendation agent that clarifies, recommends, refines, and compares options through natural dialogue instead of keyword search.
- Implemented FAISS semantic retrieval (338 catalog embeddings, MiniLM) grounded strictly to a validated product catalog, eliminating hallucinated results.
- Enforced schema compliance and guardrails (off-topic refusal, prompt-injection resistance) via FastAPI + Pydantic, validated by a 25-test automated suite.

PROJECTS

Persona Drift & Conflict Detection Engine

Python, Flask, Docker, ChromaDB, MiniLM, NLI

- Built a persona drift detector that bins conversations into day-buckets and scores sentiment (VADER) + TF-IDF topic shift to flag tone/mood change over time.
- Trained a CPU-only intent classifier (TF-IDF + Logistic Regression, 5 classes) under 50MB with 1–7ms inference latency on 191K+ weak-labeled samples.
- Built a contradiction-resolving RAG pipeline: ChromaDB + MiniLM retrieval, composite re-rank (cosine + recency + sentiment), and NLI cross-encoder scoring to surface conflicting statements in a user's history.
- Containerized the full 3-pipeline app in Docker (1.65GB, zero cold-start) and deployed a live demo on HuggingFace Spaces.

LLM Hallucination Detection & Mitigation System (HAL-RAG)

Python, NLP, RAG

- Designed a dual-stage validation pipeline that cross-references LLM outputs against a retrieval-augmented knowledge base to catch factual inconsistencies in real time.
- Analyzed linguistic uncertainty and confidence markers to flag non-factual generations at 92% precision, cutting overall hallucination rate by 40%.
- Published findings as an IEEE paper (HAL-RAG framework).

Multi-Modal Interview Performance Analyzer

Python, TensorFlow, PyTorch, Computer Vision

- Built a transformer-based feature fusion model combining text, audio, and video signals to evaluate interview performance.
- Tracked 15+ behavioral and tonal markers for sentiment and soft-skill scoring, reaching 88.7% accuracy.
- Optimized the pipeline to generate a full multi-dimensional performance scorecard in under 60 seconds; published as an IEEE paper.

PUBLICATIONS

HAL-RAG: A Retrieval-Augmented Framework for Detecting and Mitigating Hallucinations in Large Language Model Responses

A Real-Time Multi-Model System for Personality and Interview Performance Analysis Using Transformer-Based Feature Fusion